1 A flat circular coil P carrying a current produces a magnetic field. When a second coil Q is placed with its centre a distance x from the centre of coil P, as shown in Fig. 1.1, an e.m.f. V may be induced in coil Q.

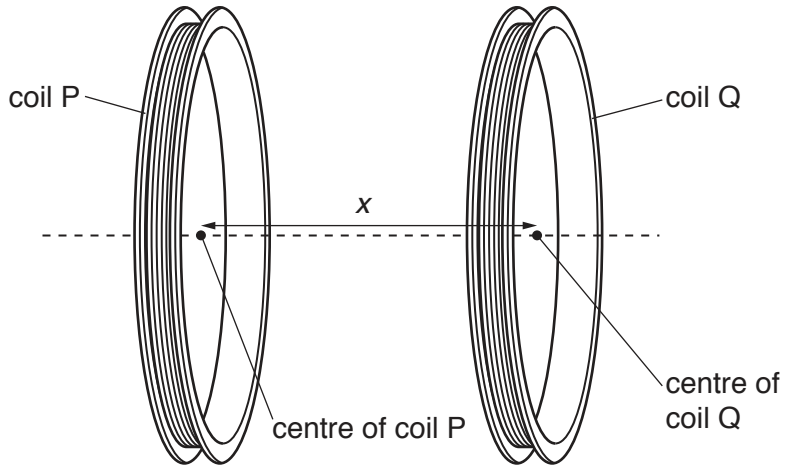


Fig. 1.1

It is suggested that V is related to x by the relationship

$$V = V_0 e^{-kx}$$

where V_0 and k are constants.

Design a laboratory experiment to test the relationship between V and x. Explain how your results could be used to determine a value for k. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

[15]

© UCLES 2017

https://xtremepape.rs/